

# Euclid’s Elements — Book II

## Proposition 4

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

*If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.*

## 1 Introduction

Proposition II.4 is the first proposition of Book II in the present reconstruction whose formal proof is naturally expressed as a composition of *two different earlier Book II propositions*. The classical theorem says that when a segment is cut into two parts, the square on the whole is equal to the two squares on the parts together with twice the rectangle contained by the parts.

The modern algebraic shadow is the familiar identity

$$(y + z)^2 = y^2 + z^2 + 2yz.$$

The formal proof, however, does not define a product of segment lengths and does not evaluate a numerical area. It remains entirely inside the rectangle and exact-scissors layer established at the beginning of Book II.

The current Lean source begins with

```
import CGJteamLab.Proposition2_2
import CGJteamLab.Proposition2_3
```

and the proof uses those imports exactly as the architecture suggests:

1. Proposition II.2 splits the square on the whole segment into two rectangles pairing the whole with the two parts;
2. Proposition II.3 expands each of those rectangles into one square and one cross rectangle;
3. representation transport identifies the two cross rectangles as two representatives of the same “rectangle contained by” the two parts;

4. addition and transitivity in `HilbertScissorsEq` combine the results.

Thus the substantial new fact at II.4 is not a new geometric construction or a new arithmetic law. It is the observation that the Book II rectangle API is already compositional enough to express the square-of-a-sum pattern using previously proved exact scissors theorems.

## 2 Euclid's Statement and Classical Configuration

### 2.1 The statement

In Heath's wording, Proposition II.4 states:

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.

Let  $AB$  be cut at  $C$ . In the geometric shorthand used throughout this volume, the statement is

$$\text{Sq}(AB) = \text{Sq}(AC) + \text{Sq}(CB) + 2 \text{Rect}(AC, CB).$$

Here  $\text{Sq}$  and  $\text{Rect}$  are expository names for geometric figures and their content. The symbol 2 means that two equal rectangle contents occur in the decomposition; it is not scalar multiplication inside the Lean segment language.

### 2.2 The classical diagram

Euclid describes the square  $ADEB$  on  $AB$  by I.46 and joins the diagonal  $BD$ . Through the cut point  $C$  he draws  $CF$  parallel to the vertical sides of the square, and through the intersection  $G$  of  $CF$  with  $BD$  he draws  $HK$  parallel to  $AB$  and  $DE$  by I.31 [?]. The square is thereby divided into four subfigures named  $AG$ ,  $CK$ ,  $HF$ , and  $GE$ .

The figure is the classical Euclidean configuration. It is deliberately not presented as the literal Lean input diagram. The formal theorem receives three configured rectangle-cut diagrams inherited from II.2 and II.3, rather than reconstructing this single classical picture point by point.

**Diagrammatic audit.** Figure 1 is retained because it is the genuine new classical state of II.4: one square, its diagonal, and the two auxiliary parallels. It should not also carry the formal proof DAG. The current Lean proof first consumes an II.2 square cut and then two independent II.3 rectangle cuts, one in the reversed orientation  $B - C - A$  and one in the direct orientation  $A - C - B$ . Those three states are therefore shown later, at the steps where the corresponding imported theorem is actually used. Pure endpoint reversal, representative rotation, addition of scissors equalities, and transitivity do not receive separate geometric figures.

## 3 Classical Proof

Euclid's proof first identifies the lower-right parallelogram  $CGKB$  as a square on  $CB$ . Since  $CF \parallel AD$ , the diagonal  $BD$  gives the relevant angle equality by I.29. The square has  $BA = AD$ ,

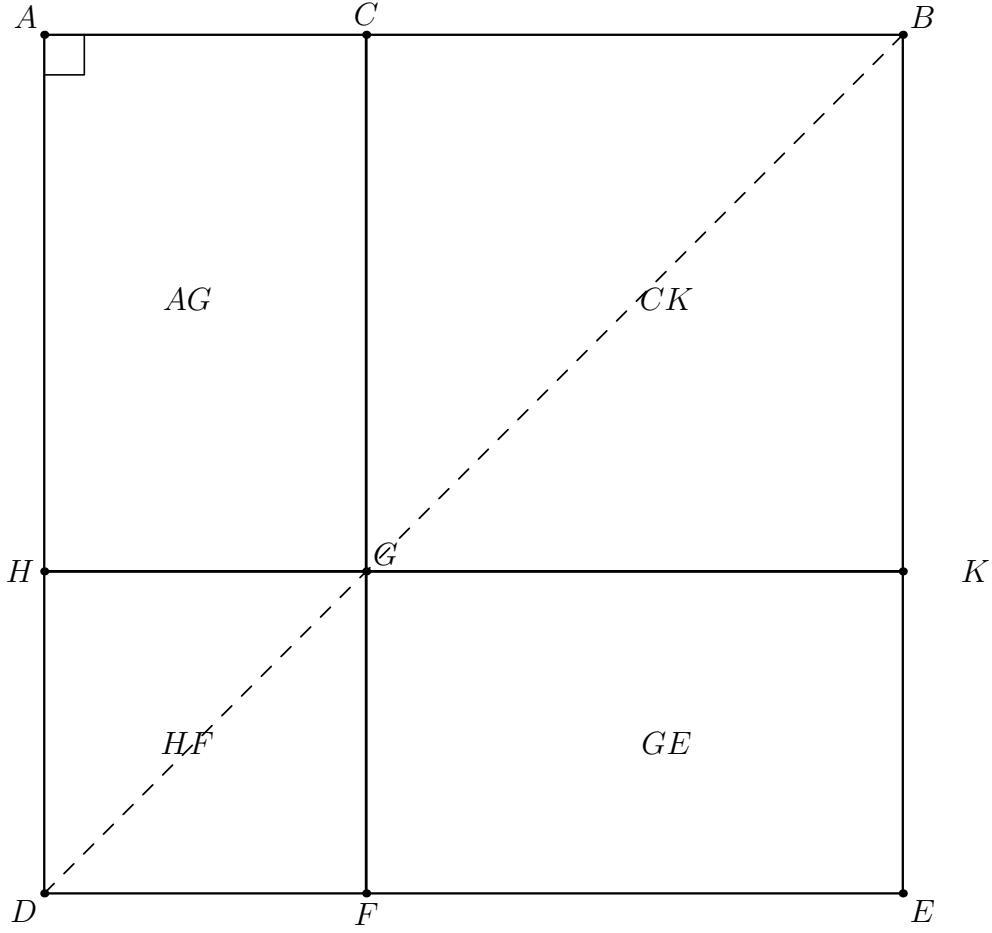


Figure 1: The classical configuration of Euclid II.4. The square  $ADEB$  is cut at  $C$  by  $CF$ , the diagonal  $BD$  meets it at  $G$ , and the parallel  $HK$  through  $G$  produces four subfigures. Euclid identifies  $HF$  and  $CK$  as the squares on  $AC$  and  $CB$ , and I.43 compares the two remaining complements  $AG$  and  $GE$ . This picture records Euclid's source geometry; it does not identify the three separate configured cut diagrams supplied to the current Lean theorem.

so I.5 gives equal base angles in triangle  $ADB$ . Combining these angle equalities and using I.6 yields

$$BC = CG.$$

Opposite sides in the surrounding parallelograms are equal by I.34, so the four sides of  $CGKB$  are equal. Right-angle information from the parallels and the square then shows that  $CGKB$  is a square on  $CB$  [?].

The same reasoning identifies  $HF$  as the square on  $AC$ . The two remaining subfigures  $AG$  and  $GE$  are complements about the diagonal of the outer square, and Proposition I.43 gives

$$AG = GE.$$

Moreover  $AG$  has adjacent side magnitudes  $AC$  and  $CG$ , while  $CG = CB$ . Hence  $AG$  is the rectangle contained by  $AC, CB$ , and  $AG + GE$  is twice that rectangle. Since the four subfigures constitute the whole square  $ADEB$ , Euclid obtains the required equality [?].

The local classical dependency graph is therefore

$$\begin{array}{c}
\text{I.46: square on } AB \quad + \quad \text{I.31: required parallels} \\
\Downarrow \\
\text{I.29} + \text{I.5} + \text{I.6: } BC = CG \\
\Downarrow \\
\text{I.34: recognize the two smaller squares} \\
\Downarrow \\
\text{I.43: complements } AG = GE \\
\Downarrow \\
\text{Sq}(AB) = \text{Sq}(AC) + \text{Sq}(CB) + 2\text{Rect}(AC, CB) \\
\Downarrow \\
\text{II.4.}
\end{array}$$

This classical DAG is important because it is *not* the DAG of the Lean proof. In particular, the formal reconstruction does not locally call I.43 and does not prove the two cross rectangles equal by diagonal complements. That equality has already been absorbed into the representative-independent rectangle calculus developed after Book I.

## 4 The Geometric Identity in the Current Formalization

The public Lean theorem starts with a strict cut

$$A - C - B.$$

It receives a square on  $AB$ , two square representatives on the two parts, and two arbitrary representatives of the cross rectangle contained by  $AC$  and  $CB$ .

At the level of exact scissors terms the final conclusion has the form

$$\text{Sq}(AB) \simeq_{\text{sc}} (\text{Rect}(AC, CB) + \text{Sq}(CA)) + (\text{Rect}(AC, CB) + \text{Sq}(CB)).$$

The use of  $CA$  rather than  $AC$  in the first square is only an endpoint orientation choice. Segments are unoriented for congruence purposes, so this is the same geometric side magnitude.

The two occurrences of the cross rectangle are deliberately represented by independent quadrilaterals. The theorem therefore says more than “the same literal rectangle occurs twice in one drawing.” It says that any two representatives satisfying

```
IsRectangleContainedBy Geo ... A C C B
```

may occupy the two cross-term positions, because the rectangle layer has already proved representative independence at the exact scissors level.

A modern mnemonic is

$$x = y + z \quad \implies \quad x^2 = y^2 + z^2 + 2yz.$$

This is useful for recognizing the combinatorial pattern only. No object corresponding to  $yz$  is constructed by `HilbertSegmentMul` in the II.4 proof.

## 5 Reusable Book II Infrastructure

### 5.1 Proposition II.2: split the square on the whole

The first reusable theorem gives

$$\text{Rect}(AB, AC) + \text{Rect}(AB, CB) \simeq_{\text{sc}} \text{Sq}(AB).$$

In the current source it is called exactly as

```

have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2

-----

-- II.2:
--
--   Rect(AB,AC) + Rect(AB,CB) = Square(AB) .
-----

have hII2 :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3 +
     hilbertParallelogramTerm Geo V0 V1 V2 V3)
    (hilbertParallelogramTerm Geo A B E0 D0) :=
  euclid_proposition_2_2
    Geo
    A B D0 E0 L0 C X0
    hSquareAB
    hACB
    hD0L0E0
    hD0X0B
    hCX0L0
    hII2LeftPar
    hII2RightPar
    U0 U1 U2 U3
    V0 V1 V2 V3
    hAB_AC
    hAB_CB

```

The final II.4 proof uses the symmetry of this relation, so the square becomes the starting point and the two whole-by-part rectangles become the two branches to be expanded.

### 5.2 Proposition II.3: expand a whole-by-part rectangle

Proposition II.3 has the generic geometric pattern

$$\text{Rect}(XY, ZY) \simeq_{\text{sc}} \text{Rect}(XZ, ZY) + \text{Sq}(ZY)$$

for a strict division point  $Z$  on  $XY$ , with the configured rectangle cut supplied explicitly.

II.4 calls this theorem twice. The right branch uses the original orientation  $A - C - B$  directly:

$$\text{Rect}(AB, CB) \simeq_{\text{sc}} \text{Rect}(AC, CB) + \text{Sq}(CB).$$

The left branch reads the same line in the reverse direction  $B - C - A$ :

$$\text{Rect}(BA, CA) \simeq_{\text{sc}} \text{Rect}(BC, CA) + \text{Sq}(CA).$$

After endpoint reversal and rectangle rotation, its cross term is another representative of  $\text{Rect}(AC, CB)$ .

### 5.3 Rectangle rotation and endpoint reversal

The only side-order normalization that is mathematically visible in II.4 is

```
rectangle_contained_by_swap
```

applied to the second cross-rectangle representative. It cyclically rotates the concrete rectangle and exchanges the roles of its containing sides while returning an exact scissors certificate for the change of presentation.

Endpoint reversal is handled by the already established congruence theorem

```
CongruentSwapSecond
```

so that  $AB$  may be read as  $BA$ ,  $AC$  as  $CA$ , and  $CB$  as  $BC$  when the reversed II.3 invocation requires that orientation. These steps are representation transport, not new metric statements.

### 5.4 Exact scissors addition

Once the two II.3 branches are available, the constructor

```
HilbertScissorsEq.add
```

combines them componentwise. Transitivity then attaches the resulting four-term expression to the square through the symmetric II.2 equality.

No subtraction, cancellation, positivity, content order, or numerical area operation occurs.

## 6 Proof Architecture of the Reconstruction

### 6.1 Step 1: reverse the cut orientation for the left branch

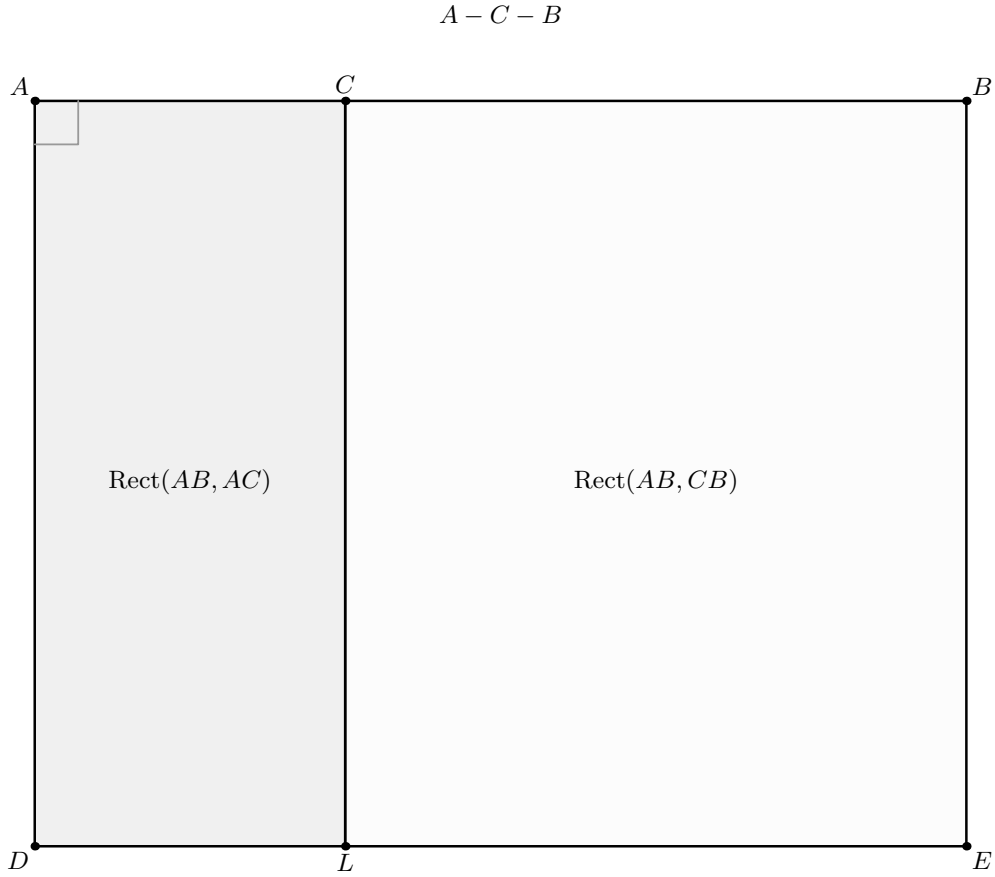
From  $A - C - B$  the proof derives  $B - C - A$ :

```
have hBCA : Geo.Between B C A :=  
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2
```

This is the only explicit betweenness reversal performed locally.

## 6.2 Step 2: apply II.2 to the square on the whole

The theorem `euclid_proposition_2_2` produces the exact decomposition of the square into the two rectangles  $\text{Rect}(AB, AC)$  and  $\text{Rect}(AB, CB)$ . Nothing about the smaller squares is used yet.



II.2 outer split

Figure 2: The first formal geometric state consumed by II.4. Proposition II.2 cuts the supplied square on  $AB$  at the strict division point  $A - C - B$  and identifies its content with two whole-by-part rectangles,  $\text{Rect}(AB, AC)$  and  $\text{Rect}(AB, CB)$ . No square on either part and no cross rectangle has entered yet. The drawing shows the dissection certified by II.2, not the arbitrary representatives  $U$  and  $V$  used by the public Lean signature.

### 6.3 Step 3: prepare the reversed left II.3 call

The first II.2 rectangle is formally contained by  $AB, AC$ , while the reversed II.3 call expects  $BA, CA$ . The same concrete rectangle is repackaged using endpoint reversal. The second final cross-rectangle representative is then rotated from the side order  $AC, CB$  to  $CB, AC$  and read, after endpoint reversal, as  $BC, CA$ .

The current source makes these transports explicit:

```
have hAB_AC_rev :
  IsRectangleContainedBy Geo
    U0 U1 U2 U3 B A C A :=
And.intro
  hAB_AC.1
  (And.intro
    (CongruentSwapSecond
      Geo U0 U1 A B hAB_AC.2.1)
    (CongruentSwapSecond
      Geo U1 U2 A C hAB_AC.2.2))

-----

-- Rotate the second cross rectangle:
--
--   Rect(AC,CB) -> Rect(CB,AC).
--
-- Endpoint reversal then reads the rotated representative as
-- Rect(BC,CA), exactly what the reversed II.3 needs.
-----

have hSwapResult :=
  rectangle_contained_by_swap
    Geo
    T20 T21 T22 T23
    A C C B
    hCross2

have hCross2Swap :=
  hSwapResult.1

have hRotateCross2 :=
  hSwapResult.2

have hCross2ForLeft :
  IsRectangleContainedBy Geo
    T21 T22 T23 T20 B C C A :=
And.intro
  hCross2Swap.1
  (And.intro
    (CongruentSwapSecond
      Geo T21 T22 C B hCross2Swap.2.1)
    (CongruentSwapSecond
      Geo T22 T23 A C hCross2Swap.2.2))
```

## 6.4 Step 4: expand the left rectangle by II.3

The reversed invocation gives

$$\text{Rect}(BA, CA) \simeq_{\text{sc}} \text{Rect}(BC, CA) + \text{Sq}(CA).$$

The rotated cross term is then transported back to the arbitrary final  $AC, CB$  representative.

The exact source block is:

```

have hLeftRaw :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3)
    (hilbertParallelogramTerm Geo T21 T22 T23 T20 +
      hilbertParallelogramTerm Geo C A F G) :=
euclid_proposition_2_3
  Geo
  B A D1 E1 L1 C X1
  hRectLeft
  hAE1_CA
  hBCA
  hD1L1E1
  hD1X1A
  hCX1L1
  hLeftLeftPar
  hLeftRightPar
  F G
  hSquareCA
  U0 U1 U2 U3
  T21 T22 T23 T20
  hAB_AC_rev
  hCross2ForLeft

-----

-- Return the left cross term to the original arbitrary
-- representative T20-T21-T22-T23.
-----

have hRotateCross2Back :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo T21 T22 T23 T20)
    (hilbertParallelogramTerm Geo T20 T21 T22 T23) :=
HilbertScissorsEq.symm
  (Geo := Geo)
  hRotateCross2

have hLeftTarget :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3)
    (hilbertParallelogramTerm Geo T20 T21 T22 T23 +
      hilbertParallelogramTerm Geo C A F G) :=
HilbertScissorsEq.trans
  (Geo := Geo)
  hLeftRaw
  (HilbertScissorsEq.add
    (Geo := Geo)
    hRotateCross2Back
    (HilbertScissorsEq.refl
      (Geo := Geo)))

```

```
((hilbertParallelogramTerm Geo C A F G)))
```

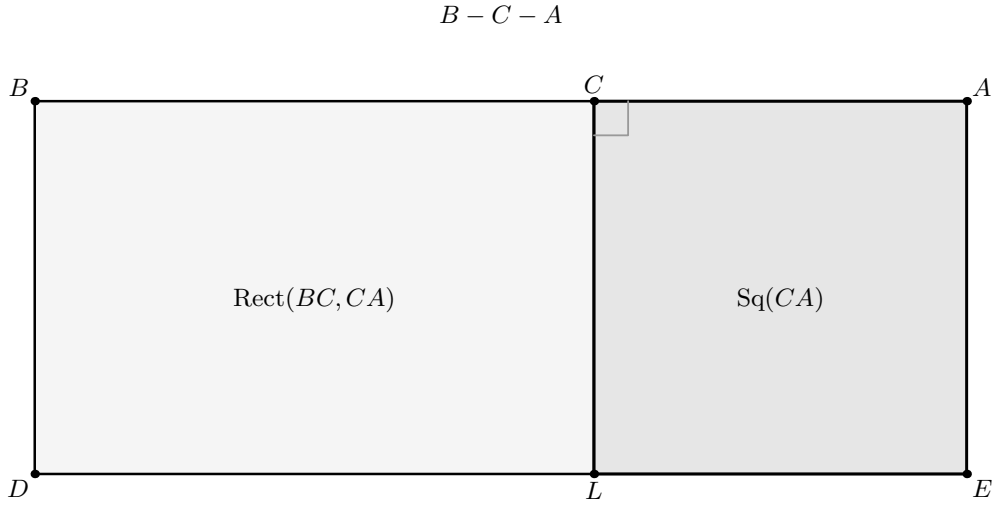


Figure 3: The left branch is II.3 read on the same divided segment in the reverse order  $B - C - A$ . The rectangle  $\text{Rect}(BA, CA)$  is cut at  $C$  into  $\text{Rect}(BC, CA)$  and the square on  $CA$ . In the Lean proof the cross term is supplied by an arbitrary representative of  $\text{Rect}(AC, CB)$ ; cyclic rectangle rotation and endpoint reversal make that representative fit the displayed  $BC, CA$  side order. Those representation transports do not change the geometric dissection and are therefore explained in the caption rather than drawn as a second state.

## 6.5 Step 5: expand the right rectangle by II.3

The right branch needs no reversal of the whole segment. It directly produces

$$\text{Rect}(AB, CB) \simeq_{\text{sc}} \text{Rect}(AC, CB) + \text{Sq}(CB).$$

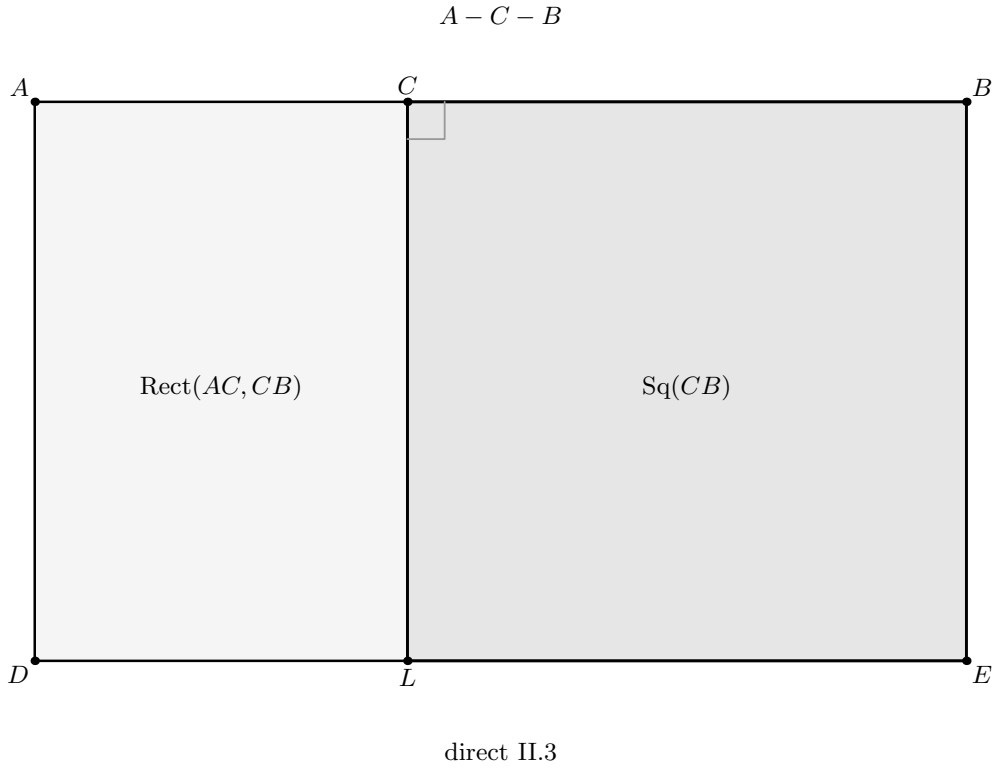


Figure 4: The second branch uses II.3 directly in the original orientation  $A - C - B$ . Cutting the supplied  $\text{Rect}(AB, CB)$  at  $C$  gives the cross rectangle  $\text{Rect}(AC, CB)$  together with the square on  $CB$ . Unlike the left branch, no reversal of the divided whole segment is required. The cross rectangle shown here is a geometric content type; the theorem permits it to be represented by a quadrilateral independent of the cross representative used in the reversed branch.

## 6.6 Step 6: add the branches and attach them to the square

The two II.3 relations are added. Finally the result is composed with the symmetric II.2 relation. The closing source is:

```
have hRight :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo V0 V1 V2 V3)
    (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
      hilbertParallelogramTerm Geo C B H K) :=
euclid_proposition_2_3
  Geo
  A B D2 E2 L2 C X2
  hRectRight
  hBE2_CB
  hACB
  hD2L2E2
  hD2X2B
  hCX2L2
  hRightLeftPar
  hRightRightPar
  H K
  hSquareCB
  V0 V1 V2 V3
  T10 T11 T12 T13
  hAB_CB
  hCross1

-----
-- Expand both II.2 summands simultaneously.
-----

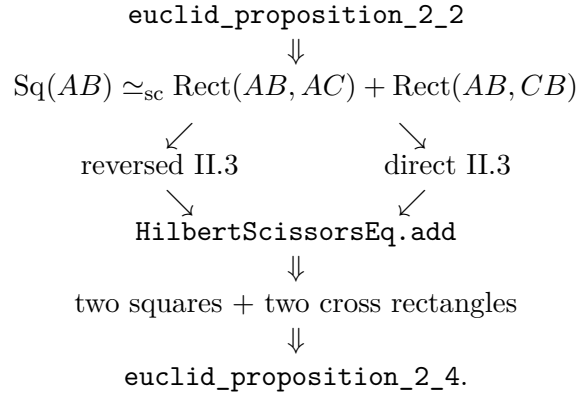
have hExpanded :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3 +
      hilbertParallelogramTerm Geo V0 V1 V2 V3)
    ((hilbertParallelogramTerm Geo T20 T21 T22 T23 +
      hilbertParallelogramTerm Geo C A F G) +
      (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
        hilbertParallelogramTerm Geo C B H K)) :=
HilbertScissorsEq.add
  (Geo := Geo)
  hLeftTarget
  hRight

-----
-- Square(AB)
--   = Rect(AB,AC) + Rect(AB,CB)
--   = the four terms of II.4.
-----

exact
  HilbertScissorsEq.trans
    (Geo := Geo)
    (HilbertScissorsEq.symm
      (Geo := Geo)
      hII2)
```

**Diagrammatic audit.** Figures 2–4 exhaust the new geometric states of the formal proof: the outer II.2 split and the two II.3 branch dissections. Step 1 merely reverses the already known betweenness  $A - C - B$ ; Step 3 repackages and cyclically rotates rectangle representatives; and Step 6 applies `HilbertScissorsEq.add` followed by symmetry and transitivity. Drawing a fourth formal picture for the final four-term sum would incorrectly suggest a newly constructed common dissection, whereas the Lean theorem intentionally allows the two cross rectangles to be independent representatives of the same contained-by data.

The actual Lean DAG is therefore



This is substantially shorter conceptually than Euclid’s local geometric DAG, but it is not an algebraic shortcut. The earlier propositions already package the relevant geometric dissections as reusable exact-scissors theorems.

## 7 Lean Formalization

The current public theorem signature is copied from the compiling `Proposition2_4.lean` source:

```

theorem euclid_proposition_2_4
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (A B C : Geo.Point)

  -----
  -- II.2 diagram: square on AB cut at C.
  -----

  (D0 E0 L0 X0 : Geo.Point)
  (hSquareAB : IsSquare Geo A B E0 D0)

  (hACB : Geo.Between A C B)
  (hD0LOE0 : Geo.Between D0 L0 E0)
  (hD0X0B : Geo.Between D0 X0 B)
  (hCX0LO : Geo.Between C X0 L0)

  (hII2LeftPar :
```

```

    IsParallelogram Geo L0 D0 A C)
(hII2RightPar :
  IsParallelogram Geo B E0 L0 C)

-- Arbitrary representatives of Rect(AB,AC) and Rect(AB,CB).
(U0 U1 U2 U3 : Geo.Point)
(V0 V1 V2 V3 : Geo.Point)

(hAB_AC :
  IsRectangleContainedBy Geo
    U0 U1 U2 U3 A B A C)

(hAB_CB :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 A B C B)

-----
-- II.3 diagram for the left term.
--
-- We read the divided whole segment in the reverse direction:
--
--   B-C-A.
--
-- Thus II.3 gives
--
--   Rect(BA,CA) = Rect(BC,CA) + Square(CA).
-----

(D1 E1 L1 X1 : Geo.Point)

(hRectLeft :
  IsRectangle Geo B A E1 D1)

(hAE1_CA :
  Geo.Congruent A E1 C A)

(hD1L1E1 : Geo.Between D1 L1 E1)
(hD1X1A : Geo.Between D1 X1 A)
(hCX1L1 : Geo.Between C X1 L1)

(hLeftLeftPar :
  IsParallelogram Geo L1 D1 B C)
(hLeftRightPar :
  IsParallelogram Geo A E1 L1 C)

-- Square on CA, i.e. on the same unoriented segment as AC.
(F G : Geo.Point)
(hSquareCA : IsSquare Geo C A F G)

-----
-- II.3 diagram for the right term:
--
--   Rect(AB,CB) = Rect(AC,CB) + Square(CB).
-----

(D2 E2 L2 X2 : Geo.Point)

(hRectRight :

```

```

    IsRectangle Geo A B E2 D2)

(hBE2_CB :
  Geo.Congruent B E2 C B)

(hD2L2E2 : Geo.Between D2 L2 E2)
(hD2X2B : Geo.Between D2 X2 B)
(hCX2L2 : Geo.Between C X2 L2)

(hRightLeftPar :
  IsParallelogram Geo L2 D2 A C)
(hRightRightPar :
  IsParallelogram Geo B E2 L2 C)

-- Square on CB.
(H K : Geo.Point)
(hSquareCB : IsSquare Geo C B H K)

-----
-- Two arbitrary representatives of the repeated cross rectangle
-- Rect(AC,CB).
-----

(T10 T11 T12 T13 : Geo.Point)
(T20 T21 T22 T23 : Geo.Point)

(hCross1 :
  IsRectangleContainedBy Geo
    T10 T11 T12 T13 A C C B)

(hCross2 :
  IsRectangleContainedBy Geo
    T20 T21 T22 T23 A C C B) :

HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo A B E0 D0)
  ((hilbertParallelogramTerm Geo T20 T21 T22 T23 +
    hilbertParallelogramTerm Geo C A F G) +
    (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
    hilbertParallelogramTerm Geo C B H K)) := by

```

Several aspects of the signature deserve explicit interpretation.

First, one strict division relation  $A - C - B$  is shared by the theorem, but the public statement receives *three* configured cut diagrams:

- the II.2 square-cut diagram, with points  $D_0, E_0, L_0, X_0$ ;
- the left II.3 rectangle-cut diagram, with points  $D_1, E_1, L_1, X_1$ ;
- the right II.3 rectangle-cut diagram, with points  $D_2, E_2, L_2, X_2$ .

The theorem does not construct these configurations from  $A, C, B$  alone. Rather, it proves the content identity once the corresponding geometric witnesses have been supplied.

Second,  $U_0U_1U_2U_3$  and  $V_0V_1V_2V_3$  are arbitrary representatives of the two rectangles produced at the II.2 level. They need not be literal subrectangles of the square  $ABE_0D_0$ .

Third, the cross terms are represented independently by  $T_{10}T_{11}T_{12}T_{13}$  and  $T_{20}T_{21}T_{22}T_{23}$ . Both satisfy the same `IsRectangleContainedBy` predicate for  $AC, CB$ . This makes the formal “twice the rectangle” statement representative-independent rather than tied to one duplicated quadrilateral.

Fourth, the conclusion is exactly

```
HilbertScissorsEq Geo ...
```

and not the weaker

```
HilbertScissorsEquicomplementtable Geo ...
```

relation. Exactness is inherited from II.2, II.3, rectangle rotation, formal addition, and transitivity.

## 8 Relationship with Earlier Book II Propositions

### 8.1 Proposition II.1

II.4 does not import `Proposition2_1` directly. Nevertheless II.1 is the underlying distributive core because both II.2 and II.3 are themselves built from its binary rectangle split. The dependency is therefore transitive:

$$\text{II.1} \longrightarrow \text{II.2} \text{ and } \text{II.3} \longrightarrow \text{II.4}.$$

This is the first point where the investment in II.1 becomes visibly compositional at the proposition level. II.2 and II.3 were sibling specializations of the binary II.1 API; II.4 uses both siblings together.

### 8.2 Proposition II.2

II.2 supplies exactly the outer decomposition needed by II.4:

$$\text{Sq}(AB) \simeq_{\text{sc}} \text{Rect}(AB, AC) + \text{Rect}(AB, CB).$$

The current II.4 source imports II.2 explicitly and calls its public theorem once.

### 8.3 Proposition II.3

II.3 supplies the expansion of each whole-by-part rectangle. The current proof calls the same public theorem twice, once after reversing the line orientation and once directly. Thus II.4 is a genuine composition theorem over the public Book II API rather than a restatement of one earlier helper.

## 9 Relationship with Book I Infrastructure

Euclid’s own local proof depends explicitly on I.46, I.31, I.29, I.5, I.6, I.34, and I.43. The formal II.4 theorem does not replay any of those arguments locally. Their geometric content has entered

the current project through the established square, parallel, parallelogram, rectangle, and scissors layers and through the already proved II.2 and II.3 theorems.

The contrast with I.43 is especially instructive. Euclid needs I.43 to prove that the two complements  $AG$  and  $GE$  in one concrete square are equal. Lean instead receives two arbitrary cross-rectangle representatives and normalizes one of them with `rectangle_contained_by_swap`. Representative independence is already built into the rectangle API, so no local complement theorem is required.

The formal proof is therefore not a transcription of Euclid’s diagram. It is a proof over a stronger reusable library whose theorems were themselves proved synthetically from the established Hilbert/Book I geometry.

## 10 Formalization Notes and Hidden Geometric Data

The polished identity hides a substantial amount of explicit positional data. The current signature contains:

- $A - C - B$ , fixing the strict order of the original segment;
- a complete square-cut configuration for the II.2 call;
- two complete rectangle-cut configurations for the two II.3 calls;
- three diagonal/cut intersection witnesses  $X_0, X_1, X_2$ ;
- six parallelogram hypotheses certifying the two pieces in each cut;
- one square on the whole and one square on each part;
- two arbitrary II.2 rectangle representatives;
- two arbitrary representatives of the repeated cross rectangle.

The proof then adds two representational obligations not visible in Euclid’s statement:

- reverse  $A - C - B$  to  $B - C - A$  for the left II.3 call;
- rotate one cross-rectangle representative and reverse segment endpoints so that the ordered side data match the reversed II.3 interface.

There is also useful negative information. No `SameRay`, `SameSide`, or `OppositeSide` datum occurs in the public II.4 proof. No new right-angle propagation theorem, no new parallel theorem, and no new scissors constructor is introduced. Those obligations are encapsulated inside the imported rectangle and Book II layers.

## 11 Geometry, Content, Representation, and Later Arithmetic

### 11.1 Geometry

The geometry consists of one segment cut  $A - C - B$ , one square on the whole, two squares on the parts, and three configured rectangle dissections. The classical source packages these ideas in one square diagram; the Lean theorem packages them as three reusable instances of already formalized cut theorems.

## 11.2 Content / scissors

The conclusion is exact `HilbertScissorsEq`. The proof only uses reflexivity, symmetry, transitivity, compatibility with addition, and exact scissors certificates supplied by II.2, II.3, and rectangle rotation. General equicomplementability is not required.

## 11.3 Representation

Representation is the genuinely nontrivial local layer. Endpoint order on the containing segments must be normalized for the reversed II.3 branch, and one cross rectangle must be cyclically rotated before it can be used in that branch. The final theorem nevertheless permits two completely independent representatives of  $\text{Rect}(AC, CB)$ .

This is the precise formal content of “twice the rectangle” in the current architecture: two scissors terms representing the same geometric rectangle magnitude occur as separate summands.

## 11.4 Later arithmetic / numerical area

Neither later layer is used. The proof imports no segment multiplication module, chooses no unit segment, proves no distributive law for a product of segment classes, and evaluates no real-valued area function.

The formula

$$(y + z)^2 = y^2 + z^2 + 2yz$$

should therefore remain explicitly labelled as the modern algebraic shadow of the synthetic construction.

# 12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore belongs to the same Euclidean project layer as II.1–II.3.

The current `Proposition2_4.lean` file introduces no proposition-local `axiom`, `sorry`, or `admit`. The theorem was compiled successfully in the project. The supplied transitive axiom audit is:

```
#print axioms Geometry.euclid_proposition_2_4

'Geometry.euclid_proposition_2_4' depends on axioms:
[propext, Classical.choice, bookZero_nullSegment1, Quot.sound]
```

Accordingly II.4 should not be described as “axiom-free.” The precise statement is that it introduces no new proposition-specific axiom and no `sorry`; its transitive dependencies are exactly the four declarations listed above. In particular the list contains no `sorryAx`.

The compile result reported for the current theorem is clean. A separate full `lake build` result was not part of the documentation input used here, so the status table below does not claim that stronger check.

## 13 Hartshorne’s Semantic Control

Hartshorne’s Section 22 supplies the appropriate general semantic control for Book II: Euclid’s equalities of plane figures can be interpreted through equal content without first assigning numerical areas [?]. He also explains why the algebraic-looking manipulations of rectangles and squares may be regarded as a kind of geometrical algebra.

II.4 is a particularly vivid example of that interpretation. Its modern shadow is a quadratic identity, but Euclid’s objects are still a large square, two smaller squares, and two rectangles. The current formalization sharpens the configured result from general equal content to exact scissors equality, because every step is built from exact dissection and exact representation transport.

Hartshorne’s “Brief Euclid” explicitly lists II.4 in the form that the square on the whole line equals the squares on its two segments plus twice the rectangle on the two segments [?]. Thus II.4 has a more direct Hartshorne-side textual counterpart than II.2 or II.3. The surrounding Section 22 discussion still supplies the essential semantic interpretation: Euclid’s equality is read as equal content, not as prior equality of numerical areas.

The stronger exact-scissors certificate remains a fact about the present Lean architecture. It should not be attributed to Hartshorne’s general content relation.

## 14 Hilbert’s Layering

Hilbert provides the complementary architectural warning. His plane-content theory and his later arithmetic of segments are distinct layers. Segment multiplication is introduced only later, using a fixed unit segment and parallel constructions, after which commutativity and distributivity become actual arithmetic theorems [?].

II.4 may look like the first place where one should invoke those later laws, because the shadow

$$(y + z)^2 = y^2 + z^2 + 2yz$$

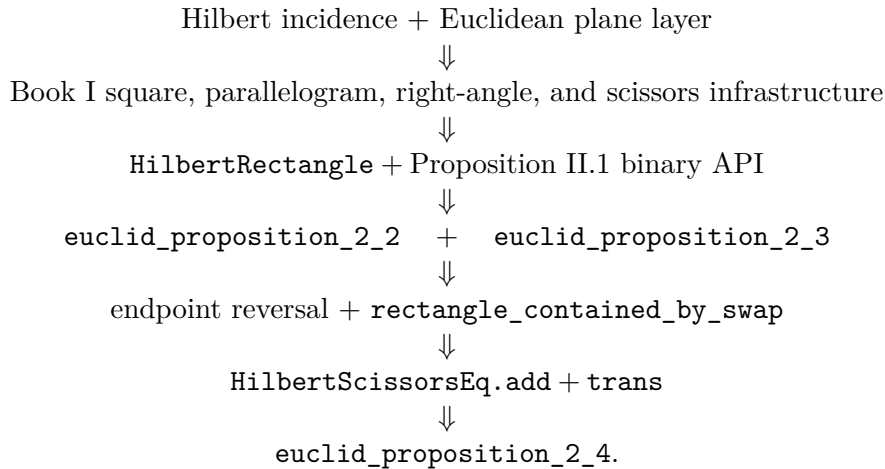
contains both distributivity and commutativity. The compiling source shows that this move is unnecessary. The proof stays below segment arithmetic:

$$\begin{array}{c} \text{rectangle/square geometry} \\ \Downarrow \\ \text{exact II.2 and II.3 scissors identities} \\ \Downarrow \\ \text{geometric representation transport} \\ \Downarrow \\ \text{exact II.4 scissors identity.} \end{array}$$

This is a useful architectural checkpoint. Even a full quadratic-looking Book II identity can be derived from the geometric rectangle calculus without crossing into Hilbert’s later segment field.

## 15 Dependency Summary

The actual Lean dependency chain is



The classical local chain is different:

$$\text{I.46} + \text{I.31} + \text{I.29} + \text{I.5} + \text{I.6} + \text{I.34} + \text{I.43} \longrightarrow \text{II.4.}$$

|                                                         |                                         |
|---------------------------------------------------------|-----------------------------------------|
| New geometric axiom in II.4:                            | no                                      |
| New content axiom in II.4:                              | no                                      |
| New proposition-local helper theorem:                   | no                                      |
| New reusable helper theorem:                            | no                                      |
| New <b>HilbertRectangle</b> theorem introduced by II.4: | no                                      |
| New Interface / Book Zero / Scissors theorem:           | no                                      |
| Reuses earlier Book II propositions:                    | yes, II.2 and II.3                      |
| Direct import of II.1:                                  | no; inherited through II.2 and II.3     |
| Direct side-order transport used:                       | yes, <b>rectangle_contained_by_swap</b> |
| Endpoint-reversal transport used:                       | yes                                     |
| Exact <b>HilbertScissorsEq</b> :                        | yes                                     |
| General equicomplementability required:                 | no                                      |
| Numerical area:                                         | no                                      |
| Segment multiplication:                                 | no                                      |
| Public Lean theorem compile:                            | yes, clean                              |
| Full relevant <b>lake build</b> :                       | not separately reported here            |
| Proposition-local <b>sorry/admit</b> :                  | no                                      |
| <b>sorryAx</b> in reported axiom list:                  | no                                      |

## 16 Conclusion

Proposition II.4 is the first strong compositional test of the Book II API. Euclid proves the theorem inside one carefully constructed square: two subfigures are shown to be squares, and I.43 identifies the two remaining complements. The current Lean proof instead works one level higher. It takes II.2 as the outer square decomposition and applies II.3 to each of the two resulting rectangles.

The only local difficulty is representational. One II.3 call uses the reversed order  $B - C - A$ , and one cross rectangle must be rotated so that its ordered containing sides match that call. Once these transports are made, `HilbertScissorsEq.add` and transitivity close the theorem.

Consequently II.4 requires no new rectangle theorem, no new scissors theorem, no new proposition-local helper, and no segment multiplication. Its apparent quadratic algebra is already present in the synthetic rectangle calculus as a composition of exact geometric decompositions. The reported axiom audit adds no new proposition-specific debt and contains no `sorryAx`.